

Can Machine Learning Models Nowcast U.S. Recessions?

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AI Usage Disclosure: I used Google Gemini for code debugging and some conceptual explanations. I hereby affirm that I did not rely on the model to generate results, choose models, or interpret my findings without first comprehensively verifying myself.

Abstract

The National Bureau of Economic Research officially dates U.S. business cycle turning points with a significant time lag, complicating economic analysis and financial decision-making by investors, businesses and policymakers alike. This research paper investigates whether machine learning algorithms can accurately nowcast U.S. recessions by analyzing contemporary macroeconomic indicators and compares them to traditional econometric benchmarks. Using “first-release” (unrevised) quarterly data for eight key economic series from 1967 to 2025, 16 classification architectures are employed, progressing from penalized logistic regression to ensemble tree methods and then neural networks. The analysis incorporates class balancing to address the scarcity of recessions vs expansions, and assesses model viability using primarily F1 score, precision and recall. The performance of these models suggest that deep learning networks capture the non-linearities of macroeconomic dynamics well; specifically, a Gated Recurrent Unit model predicts best with an F1 score of 0.89, demonstrating strong nowcasting ability without missing recessionary periods.

Introduction

Recessions are far less common than economic expansion, but macroeconomic theory tells us they are exponentially more dangerous. As the economy contracts, businesses lose money or go bankrupt, employees lose their jobs, and many may suffer before the economy rights itself. One serious problem is that recessions are almost exclusively declared on a backward-looking basis. In fact, it took almost one whole year for the National Bureau of Economic Research (NBER) to declare that the United States was in a recession starting at the end of 2007 (NBER, 2008). When the effects of a recession could be disastrous, understanding when they could occur became more important to investors, policymakers, and even central banks. Historically, economists could predict with fairly strong precision whether a recession was upcoming. However, “nowcasting”, or understanding whether the economy is currently in a recession, is newer, though important research exists in the space despite its modernity. For example, Chauvet and Piger, 2008, built a strong model relying on classical statistical models. Yet, there is comparatively less research involving machine learning models on the same topic. I intend to address this gap through 16 models evaluated in this paper, ranging from simpler penalized logistic regressions to more complex Gated Recurrent Units. I assess these models against

traditional economic models and each other. I aim to determine whether these algorithms can provide a similar or more accurate nowcast of U.S. recessions.

Motivation

Despite recessions being an important part of macroeconomic analysis, they are not particularly well defined. The common “rule of thumb” is that two consecutive quarters of negative real GDP growth define a recession, but the true definition is far more subjective. The United States has arguably the most comprehensive definition of recessions, which are defined by “turning points” – a peak and a trough in the business cycle, between which exists a recession. These turning points are decided by the National Bureau of Economic Research (NBER), but the level of analysis conducted on these turning points generally mean that dates are not released until months after the turning point, and are often revised (NBER, 2025). One critique is that this extended timeframe means that recessions have already started and ended by the time the NBER makes its decision.

To combat this lag created by NBER decision making, recessions are often predicted in advance. One of the most notable predictors involves the yield curve in financial markets. Pioneered by Estrella and Mishkin, 1995, the inversion of the yield curve shows a strong signal of a recession within the next 4-6 quarters. While this is a strong predictor, at a given time, one cannot determine if they are in the middle of a recession using the yield curve. In fact, within a recession, as rates are cut by the Fed to boost the economy, one would expect the yield curve to appear normal (low short-term rates, high long-term rates) as investors look optimistically to hold longer-term duration bonds. All this to say, by looking at a yield curve on a given day, even though the economy may truly be in a recession, the yield curve would look highly optimistic, and one could even conclude the wrong state of the economy.

This leads to the concept of “nowcasting” – assessing whether the economy is in a recession at the given moment, considering only today’s information (or including past information). The term “nowcasting” itself arose in 2008 with Giannone, Reichlin, and Small, who devised the term when predicting current-quarter GDP growth using inter-month data releases. Despite recessions existing far longer than nowcasting as a concept, there is a significant body of work which assesses the viability of nowcasting recessions. Most notably for

this paper, Chauvet and Piger devised a model in 2008 which resulted in no false positives, though it relied heavily on classic¹ econometric models like Markov Switching. Within this paper, I intend to build on this work, and other work in the macroeconomic space, by assessing whether machine learning algorithms, both basic and complex, can replicate or surpass traditional benchmarks for recession nowcasting.

Data

On its website, the NBER shares what its most common indicators are when it assesses turning points. Aside from the traditional, rule of thumb, measure of economic activity in GDP and/or GDI, they look at six other indicators:

- Employment
- Nonfarm Payrolls
- Real Personal Consumption Expenditures
- Personal Income Less Transfers
- Manufacturing and Trade Sales
- Industrial Production Index

The NBER also clarifies that there is no set methodology or importance levels regarding any of these indicators (NBER, 2025). This works out ideally for a machine learning application, as it allows the algorithm to determine the importances itself without straying far from the NBER's decision-making process.

These data series are all publicly available through FRED, and I retrieved them using the FRED API². Notably, I use the “First Release” of each data series, ignoring any revisions that were made to the numbers at a later date. The reason for this is that I believe it more closely replicates how an analyst would assess the probability of a recession in the moment: though the numbers may have been updated in 2010, an analyst would have only been able to determine a recession in 2008 using that data. Thus, I do my best to ignore any revisions to the data in my models.

¹ By “classic” I mean non-machine learning based. I do not think that Markov switching models are particularly classical or dated by any means, rather they rely on more traditional statistical methods than machine learning.

² The names of the specific data series are all cited within the References section.

As the data series are from different government entities, they all have differing timeframes of release and length of data available. To coincide with the measures of GDP and GDI, I convert all other series to quarterly measurements, also standardizing all features. This also works well with the measurement of recessions, which I detail below. The common dates for which all data has been recorded is from 1967 to 2025. I take quarterly differences to impose stationarity on the data and improve interpretability between metrics (which are measured in different magnitudes). This yields 234 observations across 8 indicators. The first 163 observations (70%) represent the training set, and the remaining are the training set. This roughly leads to a cutoff of 2008 for dividing the data.

For my measure of recessions, it is a simple binary indicator on a given observation indicating whether the NBER determined the quarter to be recessionary or not. For example, the Global Financial Crisis lead to recessions in the US from 2007Q4 to 2009Q2 (the turning points themselves were in December 2007 and June 2009) (NBER, 2023). I manually created the series, by simply creating the binary indicator variable myself, i.e., adding 1s from 2007Q4 to 2009Q2. It is worth highlighting that a good portion of the research on nowcasting (such as Chauvet and Piger, 2008) focuses mainly on the turning points. Instead, I choose to focus on whether a quarter is recessionary (Yes or No) vs the specifics of the turning points themselves. I do not believe this reduces the validity of my models, as the NBER itself tends to report recessions as quarterly metrics as well as monthly (NBER, 2023). One logical extension of my work would be to increase the frequency of observations, though a more frequent metric of aggregate production (GDP/GDI) would be needed.

Methodology and Results

I implement a total of 16 models, though many are minor variations of a larger model concept. However, to begin, it is important to establish a “traditional” baseline. For this, I extract the concepts from Chauvet and Piger and extrapolate the concept to work in comparison to my specified models later in the paper. Chauvet and Piger implement what could be described as principal component analysis, in a way, where they add a component which is common across four indicators to their data and perform their model on this component. Their four indicators are industrial production, nonfarm payrolls, personal income less transfer, and manufacturing and trade sales (which are all part of my analysis as well). Using this component, they apply a

Markov switching model, with two regimes, where the regime with a negative mean and higher variance indicated recessionary periods³. I was not able to replicate a false-positive free model as the researchers were originally, which I attribute to a slightly different dataset – the test set begins in 2008 whereas the original research was released in 2008. Still, the Chauvet-Piger (CP2008) model boasts a high accuracy and mediocre results for precision, recall, and F1 scores. It is assessed against my machine learning models in Figure 1.

The Null Model

As another baseline to compare the machine learning models to, I first fit a “Null model” which is simply predicting no recession (0) for every quarter. Interpretation-wise, this is not all that different from how the NBER currently operates – every quarter is considered to not be a recession until it is retroactively updated after the committee convenes. So, one could consider this to also be another form of a traditional baseline.

The Null model highlights one key issue with these models, which is the scarcity of recessions. The bulk of quarters are expansionary periods, meaning that the classes (0 and 1) are far from balanced. This means that by predicting all 0s, the model yields a very high accuracy, nearly 88%. To combat this, I will need to implement class balancing in other models. Further, more than one metric will be assessed for model performance. Accuracy is fairly weak as a measurement in this scenario as already outlined, and F1 score is a superior metric in my opinion. The F1 score represents how much the model “misses”, and if it misses often, the F1 score will be low. This is because the F1 score is a function of precision and recall, the other two important scoring metrics for this paper. Precision, in my scenario, represents “crying wolf” – calling a quarter a recession when it was in fact not. Recall is the opposite: one could say we are “sleeping at the wheel” or missing important recessionary quarters. The F1 score works well generally, but precision and recall together paint an important picture. A model which “cries wolf” and over-predicts recessions may cause companies to tighten up, banks to lower interest rates, etc., but if the recession does not come, there are only minor losses to the bottom line (theoretically) if the recession doesn’t materialize. By contrast, if the model is missing

³ I leveraged AI heavily to transfer these concepts to Python code, as Markov switching was outside the scope of the course and I had only a slight familiarity with the model. As per the disclosure on the cover, I did not implement or rely on any results without verifying the work.

recessions, businesses, banks and individuals have no warning and the recession hits. By contrast, a low recall can lead to businesses going bankrupt, banks going on runs, and generally a lot of harm to individuals. This is why it is important to view all metrics (including accuracy) together, and in my opinion, recall can be viewed as a tiebreaker due to the phenomena I laid out above. Because the null model has neither precision nor recall, it does not have an F1 score and generally is not very helpful for nowcasting as it has nothing to say. With this established, we can move to the next model.

Logistic Regression

A logistic regression is a logical next step in complexity. I use 10-fold time series cross validation, with a wide variety of regularization parameters (C), putting it through scikit-learn's GridSearchCV function to assess which model is the best (this is true for all below models except for the neural networks).

First, I fit a logistic regression with no penalty and balanced classes. The model selects a high level of regularization. The model accurately predicts 7 of the recessions within the test set but it does have a low level of precision, throwing a "False Positive" many times. Overall, the model has a lower accuracy than the null model and the traditional CM2008 model, and its precision and recall are also fairly low. To remedy this, I implement a penalty within the regression. First, I imposed the Ridge penalty. The model is functionally identical to the original logistic regression model, just including the L2 regularization this time. This model yields the same F1 score as no penalty, though it does increase its recall, at the cost of a lower precision. The slight improvement to recall is ideal but the F1 score indicates a model still inferior to traditional benchmarks.

Next, I impose the Lasso penalty on the regression. This model selects a smaller regularization parameter but still fails to produce an improved result over the baseline. It has the same precision as Ridge, but with a worse recall leading to an overall worse F1 score than either the standard or Ridge logistic regressions. Seeing as neither Ridge nor Lasso provided much improvement over the base, it makes sense that the Elastic Net penalty also failed to improve the over the baseline. The Elastic Net uses only 10% of the lasso component but still yields a low F1 score of 0.48, and low precision and mediocre recall scores. Regression is clearly not the ideal

model for nowcasting, indicating that the data probably has some non-linearities to consider. This also makes some intuitive economic sense, as if recessions could be nowcasted simply using a regression (which is not all that computationally intensive), one would expect significantly more usage of this sort of model within the macroeconomics space.

Tree Methods

Decision trees do notably well with non-linearities within a dataset. Further, they provide a simple and intuitive decision process which is highly interpretable for anyone, with or without macroeconomic knowledge. The classification tree I created is 3 levels deep, and highly intuitive to follow (Figure 2). In fact, its main node relies on GDP growth falling, which is in line with the “rule of thumb”. However, the standard tree’s high variance is exposed with this model, and it generalizes poorly to the test set. The classification tree has the lowest accuracy, F1 score, and precision thus far with no improvement to recall.

I move to a far more contemporaneous model, the Random Forest classifier. Again, I let the model determine its own optimal tree depth, number of estimators, and samples per leaf. It decides that trees are maximum 3 levels deep (much like the standard tree) and creates 50 of them. With this configuration, the Random Forest classifier returns the best F1 score, accuracy, precision and recall so far, with an F1 score near 0.73 and recall nearing 0.90.

I apply a similar process with a Gradient Boosted classifier. However, a technical limitation arises here: the Gradient Boosted function within scikit-learn does not allow for balancing of classes as has been done for every model thus far. So, I run the model regardless, but I consider the model results to be more of an indicator for future models, as opposed to a primary result of this research⁴. The Gradient Boosted trees yield incredibly strong results, with an F1 score of 0.82 and high precision, though recall does fall relative to the Random Forest regressor. XGBoosting does not have the same limitation, so I run this model with full confidence and come to a slightly worse F1 score than either of the other ensemble tree methods. However, it boasts a high recall unlike standard Gradient Boosting. All in all, ensemble tree

⁴ I believe more complex/comprehensive gradient boosting libraries exist, but I am unaware of their implementation. There is certainly potential for future research, or to revisit if GradientBoostingClassifier is updated in scikit-learn.

methods provide a significantly stronger model than other models thus far, showing promise for more complex models down the line.

Support Vector Machines

Support Vector Machines seem, in theory, viable for this problem. It is a classification problem with a (relatively) small dataset which is generally where SVMs excel. I test SVMs with both the linear and RBF kernel. The linear kernel, somewhat expectedly, does not perform all that well, with an F1 score of 0.45 and low precision. It does have a fairly strong recall, but at the cost of all other metrics. The RBF kernel slightly improves the accuracy and precision of the model but at a hit to recall and F1 score (slightly). It is possible that an SVM with polynomial terms and/or a different kernel could improve these results, but considering their results relative to the tree methods, I do not consider them to be a strong model option for this research topic.

Neural Networks

Given the non-linearities of this problem, one could expect neural networks to perform well. However, it is important that they are specified well for the problem. I create a variety of neural networks to assess different fits.

The first network is the most basic, “vanilla” neural network. It is a 2-layer neural network, with hidden layers of 16 and 8 neurons. 16 is optimal I believe, as any higher and the number of parameters will exceed the number of quarters. With a small amount of training data (~160 quarters), too many neurons will lead to the network simply memorizing the data. With 32 neurons, the number of parameters would be $8 \times 32 = 256$, far exceeding the number of quarters available to train and running the risk of overfitting the model. 16 seems to strike a nicer balance at $8 \times 16 = 128$ parameters which seems more fitting. The data passes through the 16 and 8 neuron hidden layers using ReLU activation and 20% dropout (due to our small training data size), then are output through a Sigmoid activation function to place the results within a 0 and 1 range of probabilities. I consider any probability >0.5 to be a prediction for a recession. It is important to note here that this could easily be adjusted to better reflect the analyst’s preferences. Perhaps they would like to be more stringent and only predict a recession with a probability of 65% or higher. Or, perhaps the analyst considers 40% a high enough probability to make their

decision. This is highly adjustable according to the need of the specific user. I compile this neural network (and all subsequent) with the Adam optimizer and early stopping. The vanilla neural network yields a strong F1 score of 0.74 and shows high results for both precision and recall which no model had previously done.

Next, I leverage the time series nature of the data and create a Recurrent Neural Network (RNN). This network is more basic than the vanilla, only 1 layer with 8 neurons. The lookback period is 4 quarters (1 year) which seemed reasonable and limited the number of lost observations. The RNN also uses 20% dropout as with the vanilla NN to combat the low number of observations. I also use L2 regularization since in the logistic regression, it provided gains over a standard regression. The RNN is then fed through the same output layer and compiled just as in the vanilla NN. The RNN demonstrates an improvement over the standard NN with a high F1 score of 0.8, as well as high precision and recall scores. The exact same methodology is copied over to the most complex models I implement, those being the Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) models.

These models are functionally different but implemented identically to the RNN. The LSTM performs just as well as the simple RNN, with slightly higher accuracy but the same F1 score, precision and recall. The GRU boasts the highest F1 score of all models at 0.89, with precision and recall also being very high. All neural networks, but specifically the GRU, far exceed the baseline models, and even the ensemble tree methods. They are the strongest models as shown in Figure 1.

Conclusion

My models demonstrate that there is potential for machine learning models to outperform traditional macroeconomic benchmarks. Though, going beyond statistical prowess, in Figure 3, I demonstrate exactly when each model is right, and when it is wrong, compared to the actual NBER classification for that quarter. My research seems to indicate that neural networks, particularly complex ones such as GRU, perform best for nowcasting the state of the economy, and notably, do not miss often i.e., “fall asleep at the wheel”. While my research does have some faults and could be improved upon, I do believe that this exists as a solid proof-of-concept for

using machine learning in recession nowcasting, and this could be built on to create a far more robust recession nowcast model down the line.

Tables and Figures

Figure 1: Model Performance Metrics

Model	Accuracy	Precision	Recall	F1 Score
GRU	0.9718	0.8889	0.8889	0.8889
GradientBoost	0.9577	0.8750	0.7778	0.8235
RNN	0.9437	0.7273	0.8889	0.8000
LSTM	0.9437	0.7273	0.8889	0.8000
NN	0.9296	0.7000	0.7778	0.7368
RandomForest	0.9155	0.6154	0.8889	0.7273
XGBoost	0.8873	0.5333	0.8889	0.6667
CP2008	0.9014	0.6250	0.5556	0.5882
Null	0.8732			
LogReg	0.8028	0.3684	0.7778	0.5000
Ridge	0.7746	0.3478	0.8889	0.5000
Lasso	0.7887	0.3500	0.7778	0.4828
ElasticNet	0.7887	0.3500	0.7778	0.4828
SVM-L	0.7606	0.3182	0.7778	0.4516
SVM-R	0.7887	0.3333	0.6667	0.4444
ClassTree	0.7324	0.2917	0.7778	0.4242

Figure 2: Decision Tree Classifier

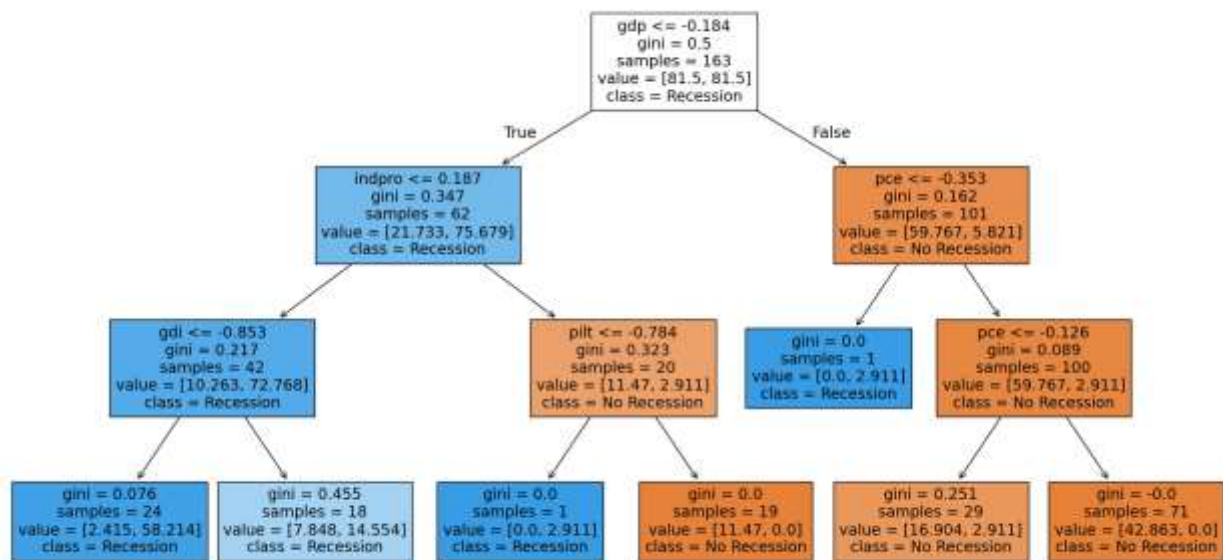
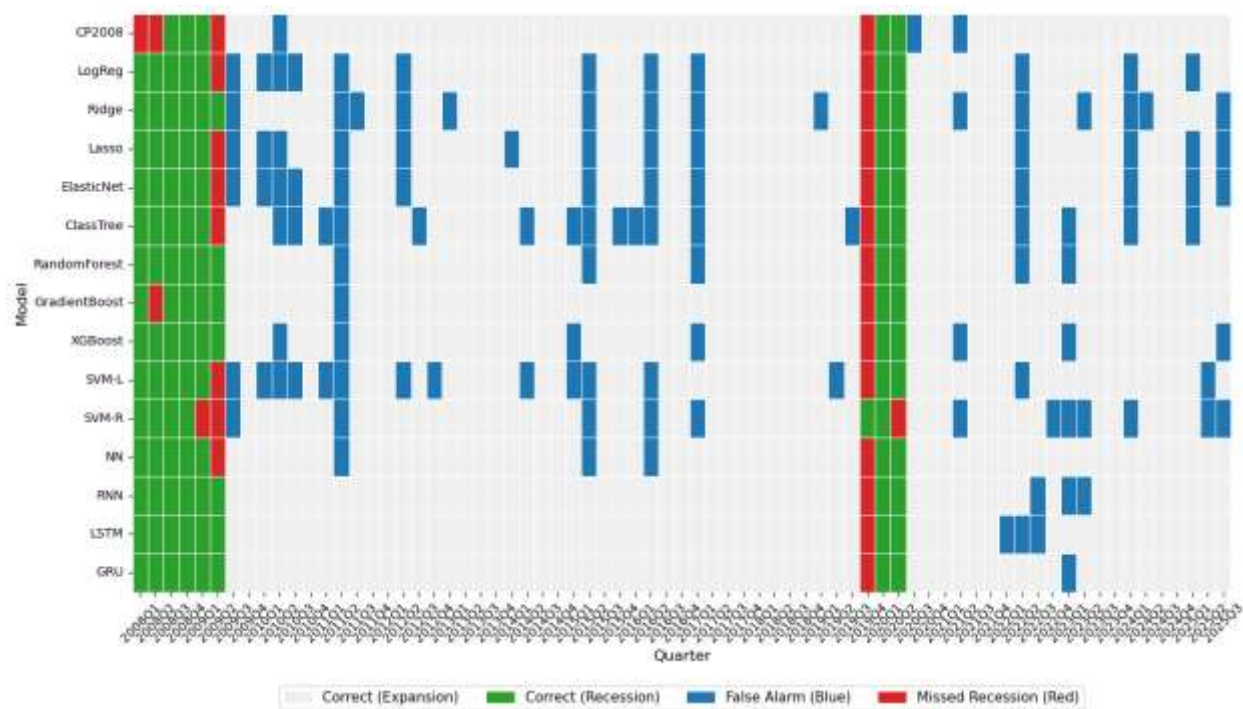


Figure 3: Model Predictions, by Quarter



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